

Table of Contents

Introduction:	1
Review of Literature:	2
Aim and Objectives:	2
Materials and Methods:	2
Results:	2
Discussion:	3
Conclusion:	3
Limitations:	3
References:	4

Pre-operative Hypoalbuminemia as a Predictor of Post operative Outcomes Following Emergency Exploratory Laparotomy: A Prospective Observational Study

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Abstract :

Background: The Emergency exploratory laparotomy is linked to considerable post-operative morbidity and mortality. Pre-operative serum albumin is regarded as a significant indicator of nutritional and physiological condition.

Aim: To assess the correlation between pre-operative serum albumin concentrations & post operative results in patients having emergency exploratory laparotomy.

Methods: A prospective observational study was performed in the Department of General Surgery at MMIMSR, Mullana, Ambala, over a duration of 18 months. Forty individuals receiving emergency exploratory laparotomy for intestinal disorders were included. Serum albumin levels were assessed pre-operatively, and patients were monitored for 30 days following the operation. The outcomes examined encompassed surgical site infection (SSI), wound dehiscence, ileus, anastomotic leak, enterocutaneous fistula, pleural effusion, intensive care unit (ICU) duration hospital length of stay, morbidity, mortality.

Results: Hypoalbuminemia (<3.5 g/dL) was detected in 87.5% of individuals. Notable correlations were identified between the hypo-albuminemia and extended ICU duration ($p=0.038$), surgical site infection (SSI) ($p=0.027$), and wound dehiscence ($p=0.034$), pleural effusion ($p=0.038$), and longer hospital duration ($p=0.001$). Mortality was confined solely to hypoalbuminemic individuals. Nonetheless, correlations with post operative ileus, anastomotic leak, enterocutaneous fistula, and ascites were not statistically significant.

Conclusion: The Pre-operative hypoalbuminemia is a significant predictor of unfavorable post-operative outcomes after emergency exploratory laparotomy. Regular evaluation of serum albumin may assist in identifying high-risk individuals and enhancing perioperative care.

Keywords: Hypoalbuminemia, surgical site infection, exploratory laparotomy, serum albumin, postoperative sequelae, and emergency surgery.

Introduction:

Emergency exploratory laparotomy is one of the most frequently executed emergency surgical interventions and is linked to significant morbidity & mortality. Delayed presentation, infection, malnutrition & physiological deterioration adversely affect post operative results. Multiple factors affect post operative outcomes after emergency laparotomy, including as age, comorbidities, and underlying pathology, symptom duration, and intraoperative contamination, and

physiological reserve. Nutritional state is the essential in influencing surgical results. Malnutrition is acknowledged as a significant factor contributing to the delayed wound healing, compromised immunological response, heightened vulnerability to infection, and extended hospitalization, and elevated mortality rates. Serum albumin is a cost-effective and readily accessible biomarker indicative of nutritional and inflammatory conditions.

Hypoalbuminemia is linked to compromised wound healing, heightened vulnerability to infection, extended hospital stays, and higher mortality. This study sought to assess the correlation between pre-operative blood albumin levels and post-operative outcomes in patients having emergency exploratory laparotomy.

Review of Literature:

- Bhate et al. (2026) conducted a prospective observational study titled "Pre-operative Serum Albumin and Surgical Outcomes in Emergency Abdominal Surgeries," using a sample size of 200 participants. The authors determined that pre-operative hypoalbuminemia is a robust, independent predictor of morbidity, death, and extended hospital stay in patients undergoing emergency abdominal surgery. Regular evaluation of serum albumin can function as a straight forward, and economical method for early risk identification and perioperative management. The study's follow-up was restricted to 30 days post-operation, hence long-term problems were not evaluated.

- Mohammad Riyaz et al. (2025) conducted a retrospective observational study with the 50 patients to evaluate the prognostic significance of pre-operative blood albumin levels concerning abdominal wound complications after emergency exploratory laparotomy. Their research indicated that the diminished pre-operative albumin levels were significantly correlated with an increased occurrence of wound related problems. Patients exhibiting serum albumin levels below 3.5 g/dL demonstrated a significantly elevated risk of surgical site infection, and wound dehiscence, and protracted wound healing in comparison to those with normal albumin concentrations. The results indicated that serum albumin is an independent factor influencing wound-related morbidity in emergency abdominal operations.

- Naga Harsha Vardhan et al. (2025) conducted a study including 84 patients undergoing major open abdominal operations to ascertain the correlation between post-operative serum albumin levels and surgical problems. Their findings demonstrated that an early decline in blood albumin levels post-surgery was strongly associated with the onset of postoperative problems. The authors highlighted that continuous monitoring of albumin levels offers a straightforward, non-invasive, and economical method for identifying patients at risk of problems. Nonetheless, the study recognized specific limitations, such as its single-center design and somewhat small sample size. The authors suggested that future studies should involve multicenter collaboration & include additional inflammatory indicators, such as C-reactive protein or procalcitonin, to enhance predictive accuracy.

Aim and Objectives:

1. To ascertain preoperative serum albumin concentrations in patients undergoing emergency exploratory laparotomy.
2. To evaluate the occurrence of postoperative sequelae, including surgical site infection, ileus, anastomotic leak, enterocutaneous fistula, and 30-day morbidity in patients with hypoalbuminemia.
3. To ascertain the prevalence of pleural effusion and ascites in hypoalbuminemic patients having emergency exploratory laparotomy.

Materials and Methods:

This prospective observational study was carried out in the Department of General Surgery at MMIMSR, Mullana, Ambala, during an 18-month duration. Forty patients consented to participate in an emergency exploratory laparotomy for intestinal diseases. The Inclusion criteria encompassed all patients receiving emergency exploratory laparotomy for intestinal conditions. The exclusion criteria encompassed immunocompromised individuals, those receiving laparoscopic surgery or mini-laparotomy, patients with significant comorbidities, and individuals requiring re-exploratory laparotomy within 30 days. A comprehensive history, clinical assessment, laboratory evaluations, ASA classification, & pre-operative serum albumin measurement were documented. Patients were monitored for 30 days post-operatively for complications such as surgical site infection, wound dehiscence, ileus, and anastomotic leak, enterocutaneous fistula, ascites, pleural effusion, intensive care unit stay, morbidity, and death. Statistical analysis was conducted utilizing SPSS version 21. The Chi-square test and Pearson correlation coefficient were employed. A p-value of less than 0.05 was deemed statistically significant.

Results:

Of the 40 cases, 75% were male and 25% were female. the Intestinal perforation constituted the predominant indication for surgical intervention (80%). The average serum albumin concentration was 2.95 ± 0.67 g/dL. Hypoalbuminemia (<3.5 g/dL) was observed in 35 individuals (87.5%). Patients exhibiting hypoalbuminemia experienced markedly elevated rates of extended ICU duration (42.9%), surgical site infections (74.3%), wound dehiscence (71.4%), and pleural effusion (68.6%). A statistically significant inverse connection was identified between serum albumin levels and the duration of hospital stay ($r = -0.58$, $p=0.001$). Mortality was noted exclusively in patients with hypoalbuminemia.

Discussion:

This study's findings indicate that pre-operative hypoalbuminemia is significantly linked to negative post-operative outcomes after emergency exploratory laparotomy. Hypoalbuminemia hinders collagen production, tissue repair, immunological function, and physiological reserve, thereby increasing the risk of problems for patients. The findings align with earlier research conducted by Kashyap et al., Bhate et al., and Hennessey et al., and which recognized hypoalbuminemia as an independent predictor of morbidity, surgical site infection, extended hospitalization, & mortality. Routine pre-operative serum albumin assessment can identify high-risk surgical patients and inform nutritional optimization and perioperative care measures.

Conclusion:

Pre-operative serum of albumin serves as a significant predictor of post-operative problems following emergency exploratory laparotomy. Hypoalbuminemia is markedly correlated with extended ICU duration, surgical site infections, wound dehiscence, and pleural effusion, prolonged hospitalization, and mortality. Serum albumin measurement should be integrated into the standard perioperative risk evaluation techniques.

Limitations:

The research was constrained by its restricted sample size and monocentric design. Follow-up exceeding 30 days was not conducted.

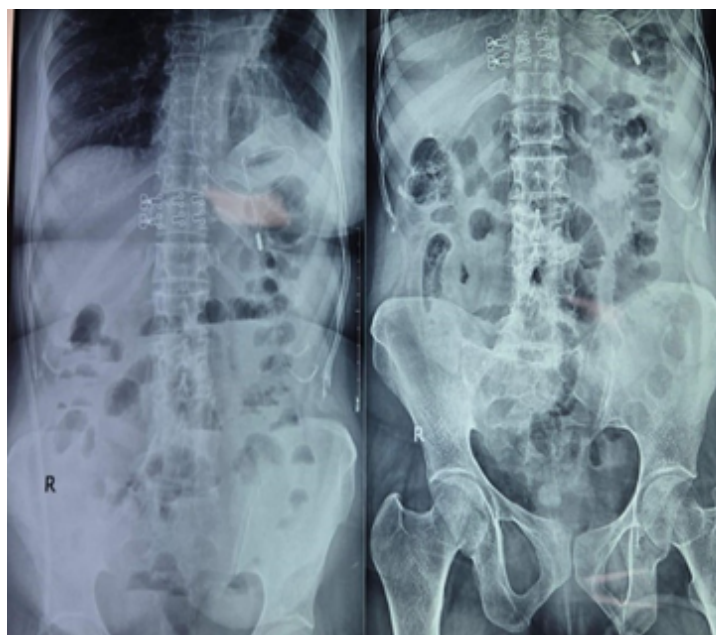


Figure-1: Multiple air fluid levels

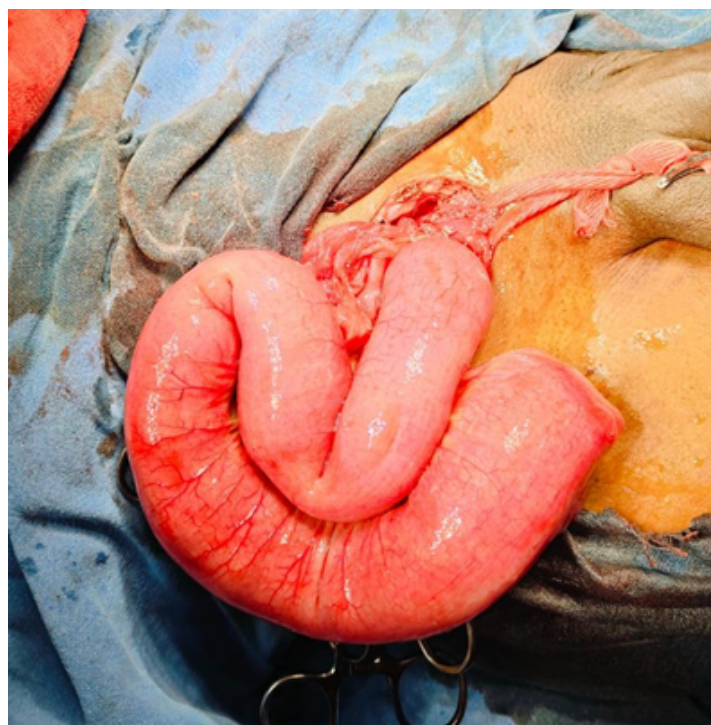


Figure-2: End ileostomy



Fig-3: Resected Gangrenous Bowel segment



Figure-4: End ileostomy



Figure-5: SSI with wound dehiscence



Figure-6: Enterocutaneous fistula

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